Inference often combines conditioning and summarizing

Observed y $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y) \, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$

Choose a reconstruction by its conditional squared-error risk

Choose a reconstruction after observing y

$$m = E[z|y]$$

Unknown $z \in \mathbb{R}^d$; choose $a \in \mathbb{R}^d$ to reconstruct it.

Squared-error criterion

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center the error at the conditional mean

$$r = z - m, \quad z - a = r + (m - a)$$

$$\text{align(center)}[\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2]$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, a and m are fixed; the residual r remains random.

Centering the error makes the posterior mean optimal

Average the centered expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

$$E[r|y] = E[z|y] - m = 0$$

$$E\big[r^T(m-a)|y\big] = E[r|y]^T(m-a) = 0$$

With $V = \mathrm{Cov}[z|y] = E[rr^T|y]$

$$E\big[\|r\|^2|y\big] = \sum_{j=1}^{d} E\big[r_j^2|y\big] = \operatorname{tr} V$$

Conditional squared-error risk

$$R(a|y) = \operatorname{tr} V + \|m-a\|^2$$

Optimal reconstruction

$$\|m-a\|^2 \geq 0 \quad \text{with equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y], \quad \min_a R(a|y) = \operatorname{tr} V$$

align(center)[For the conditional Gaussian: $\quad a^*(y) = \mu_{z|y}, \quad \min_a R(a|y) = \operatorname{tr} \Sigma_{z|y}$]

- Risk is residual uncertainty plus the squared distance of the chosen estimate from the mean.
- The conditional mean minimizes model expected squared error.

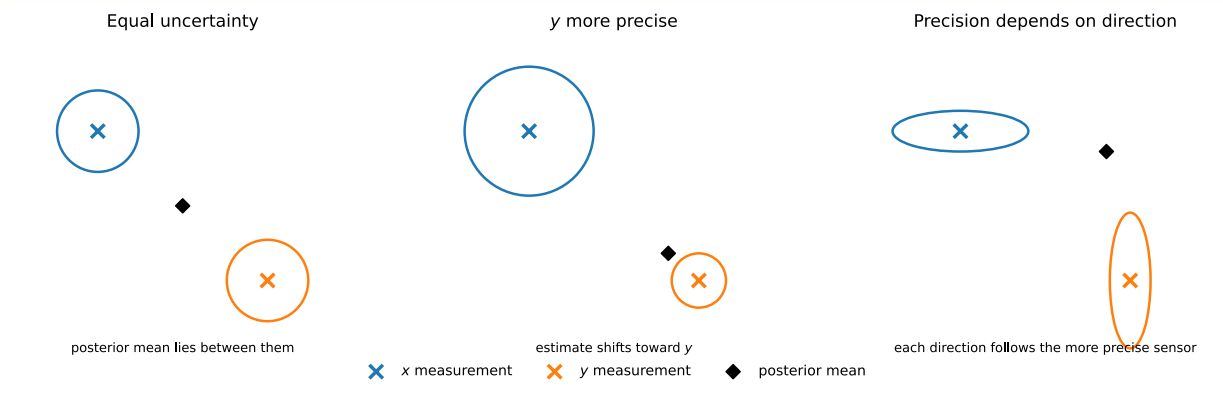
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction